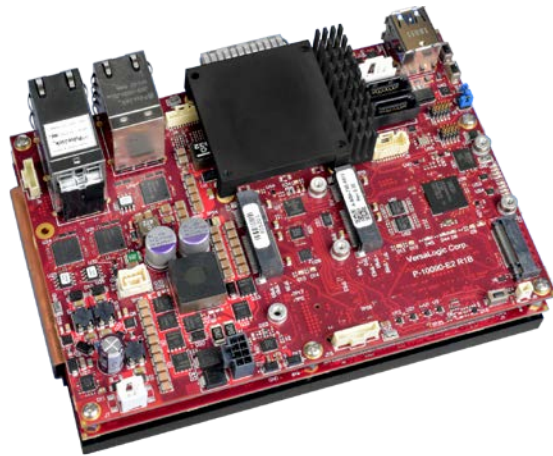




Remote Management Users Manual

REV. February 2020

Grizzly (VL-ESU-5070)

A rugged embedded server unit featuring the Intel® C3958 16-core micro-server processor and 10 GbE networking ports.





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Product Release Notes

Rev 1.00 – Initial Release

Support Page

The [Product Support Page](#) contains additional information and resources for this product including:

- Operating system information and links to software drivers
- Data sheets and manufacturers' links for chips used in this product
- BIOS and PLD information and upgrades
- Links to Knowledge Base articles and product advisories

VersaTech Knowledgebase

The [VersaTech KnowledgeBase](#) is a useful resource for resolving technical issues with your VersaLogic product.

Customer Support

If you are unable to solve a problem after reading this manual, visiting the product support page, or searching the Knowledge Base, contact VersaLogic Technical Support at (503) 747-2261. VersaLogic support engineers are also available via e-mail at Support@VersaLogic.com.

Repair Service

If your product requires service, you must obtain a Returned Material Authorization (RMA) number by calling (503) 747-2261. Please provide the following information:

- Your name, the name of your company, your phone number, and e-mail address
- The name of a technician or engineer that can be contacted if any questions arise
- The quantity of items being returned
- The model and serial number (barcode) of each item
- A detailed description of the problem
- Steps you have taken to resolve or recreate the problem
- The return shipping address

AS9100

All AS9100 products dispositioned for scrap shall be conspicuously and permanently marked, or positively controlled, until physically rendered unusable.

Material designated for scrap may be recycled in a manner that complies with applicable environmental regulations.

Note:

VersaLogic recommends that all materials be disposed of in an environmentally responsible manner i.e., recycling in compliance with applicable laws and regulations.

Warranty Repair

All parts and labor charges are covered, including return shipping charges for Standard Ground delivery to United States addresses.

Non-warranty Repair

All approved non-warranty repairs are subject to diagnosis and labor charges, parts charges and return shipping fees. Please specify the shipping method you prefer and provide a purchase order number for invoicing the repair.

Note:

Please mark the RMA number clearly on the outside of the box before returning.

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Introduction

Description

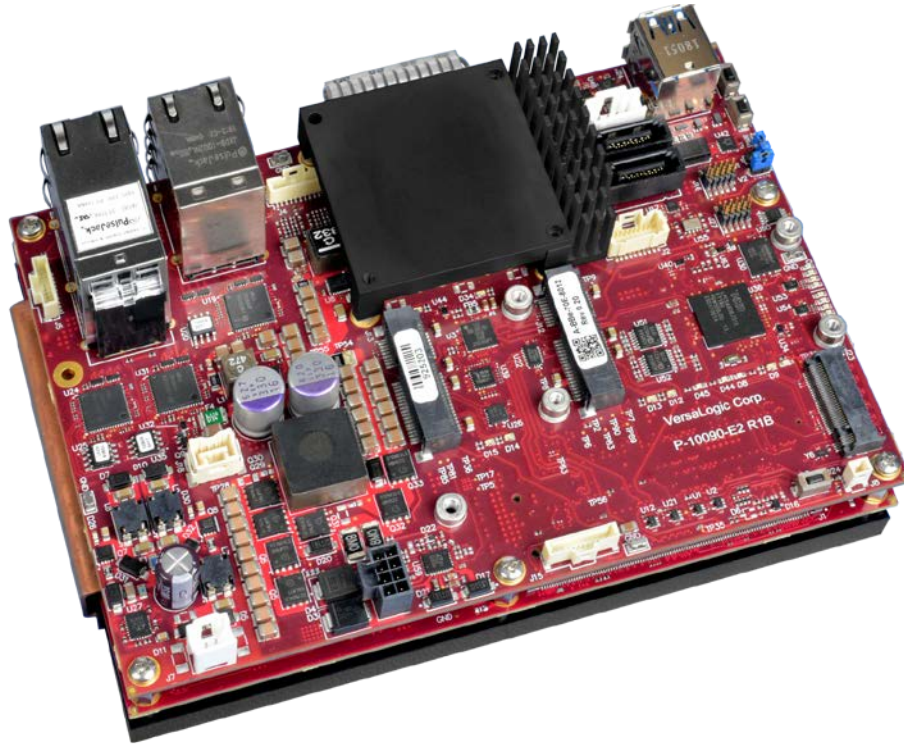
The VL-ESU-5070 is a rugged embedded server unit (ESU) featuring an Intel 16-core processor, two 10 Gigabit Ethernet SFP+ ports, four Gigabit Ethernet ports, and up to 128 GB of ECC memory. This configuration is ideal for applications requiring high-performance processing using high data bandwidth. Additionally, two Mini PCIe sockets and a PCIe x4 M.2 site provide on-board I/O expansion and high speed read/write high-capacity storage. The VL-ESU-5070 also contains additional interfaces including USB, serial and digital I/O, and SATA.

The high-performance capabilities of the VL-ESU-5070 make it ideal for situations where local data gathering and processing are required for security or location-based cloud computing. A 16-core processor coupled with ECC memory enables the use of hypervisors supporting virtual machines. The 10 Gigabit SFP+ ports permit high speed connectivity. Networks can be created using plug-in copper, short-reach fiber, or long-reach fiber transceivers.

- **Intel® 16-core server-class processor**
Very high performance computing and I/O processing
- **-40° to +85°C Operation**
Operates over full industrial temperature range
- **MIL-STD-202H Shock and vibration**
Ideal for Mil/Aero and other challenging environments
- **Error correcting RAM safeguards critical applications**
Four slots support up to 128 GB of ECC memory
- **Two 10 Gigabit Ethernet (SFP+)**
Supports very high speed copper and fiber connections
- **Four 1 GbE Ports. Two with Power Over Ethernet (POE)**
- **Mini PCIe Sockets. Add GPS and other options**
- **On-board data storage**
M.2 expansion site supports up to 2 TB of storage
- **Compact size. Only 110 x 155 mm (4.33 x 6.1")**

Introduction

VL-ESU-5070 boards undergo 100% functional testing and are backed by a limited five-year warranty. Careful parts sourcing ensure the highest possible quality, reliability, and product longevity for this exceptional embedded server. A US-based technical support staff ensures quick, expert assistance.



Configuration and Setup

Initial Configuration

The following components are recommended for a typical development system.

Configuration with Video

- VL-ESU-5070 computer
- VL-CBR-0812 Power cable, 10 to 30V
- MPEe-V5E Mini PCIe Video Expansion Module (installed at J12)
- VL-CBR-1204 VGA Cable for MPEe-V5E
- VL-CBR-0203 Battery (J8)
- SATA hard drive
- VL-CBR-0702 SATA cable
- VL-CBR-0407 SATA Power cable
- USB keyboard and mouse

Configuration without Video

- VL-ESU-5070 computer
- VL-CBR-0812 Power cable, 10 to 30V
- CBR-0203 Battery (J8)
- Ethernet connection to host
- USB keyboard and mouse

Note: Air flow over the system is required for setup and configuration

You will also need an operating system installation media.

Basic Setup

The following steps outline the procedure for setting up a typical development system. The board should be handled at an ESD workstation or while wearing a grounded antistatic wrist strap.

Before you begin, unpack the product and accessories. Verify that you received all the items you ordered. Inspect the system visually for any damage that may have occurred in shipping. Contact Support@VersaLogic.com immediately if any items are damaged or missing.

Gather all the peripheral devices you plan to attach to the board and their interface and power cables.

It is recommended that you attach standoffs to the board.

1. Attach Cables and Peripherals

- Install MPEe-V5E Mini PCIe Video Expansion Module (J12)
- Install VL-CBR-1204 VGA Cable from the MPEe-V5E card to monitor
- Install VL-CBR-0203 Battery (J8)
- Install keyboard and mouse
- Attach SATA hard drive using the VL-CBR-0702 SATA cable and VL-CBR-0407 SATA Power cable

Note: Model VL-ESU-5070ECP-00 will require memory to be installed

2. Attach Power

- VL-CBR-0812 Power cable to power connector (J6)

3. Review Configuration

- Before you power up the system, double check all the connections. Make sure all cables are oriented correctly and that adequate power will be supplied to the board and peripheral devices.

4. Power On

- Turn on the power supply and the display. If the system is correctly configured, a video signal should be present.

5. Select a Boot Drive

6. Install Operating System

- Install the operating system according to the instructions provided by the OS manufacturer. (See Operating System Installation.)

Note: Initial boot time will be 5-10 minutes.

Operating System Installation

The standard PC architecture used on the VL-ESU-5070 make the installation and use of most of the standard x86 processor-based operating systems very simple. The operating systems listed on the VersaLogic Software Page use the standard installation procedures provided by the maker of the OS. Special optimized hardware drivers for a particular operating system, or a link to the drivers, are available at the VL-ESU-5070 Support Page.

Remote Management Summary

Remote Management on the ESU-5070 (Grizzly) platform is an IPMI 2.0 compliant solution implemented in the Innovation Engine of the Intel Denverton (Atom C3xxx) SoC. The Innovation Engine (IE) is a 32-bit Intel Quark-based processor core that, along with firmware, serves the same function as a dedicated Baseboard Management Controller (BMC) without the need for additional components on the baseboard.

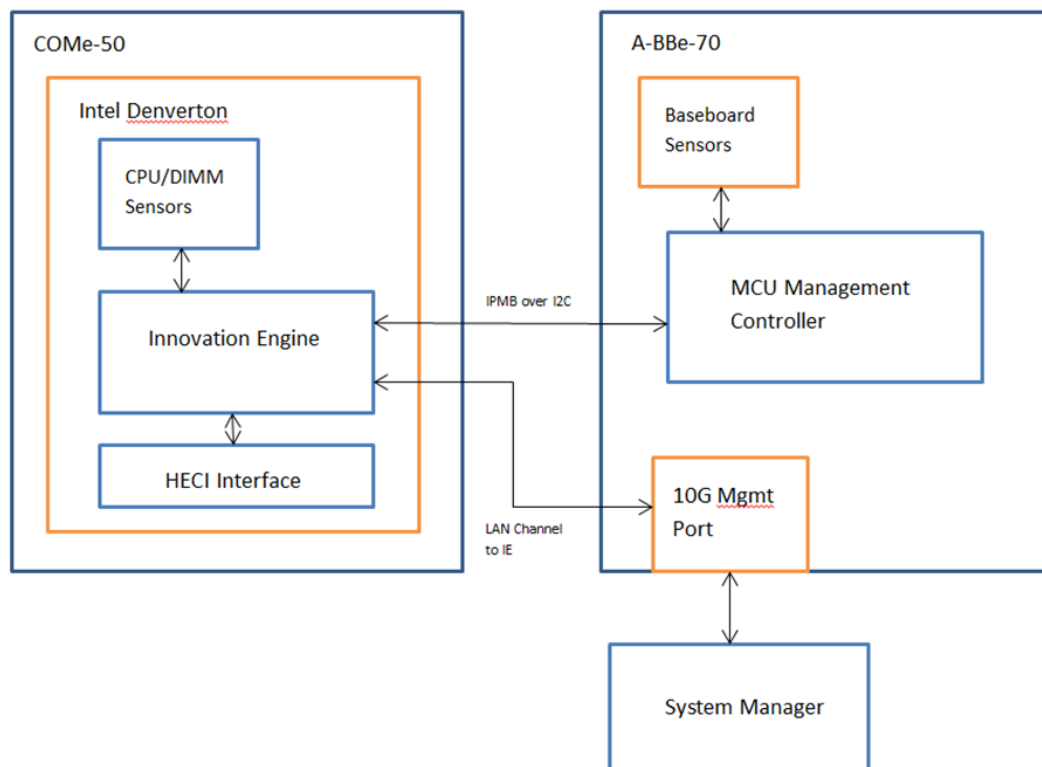
The IE can be accessed through a LAN or a HECI interface using a customized version of the ipmitool utility. IPMI commands compose the interface to the Remote Management on Grizzly. There are many methods to send IPMI commands and many open source utilities available, but Versallogic recommends ipmitool. The IPMI LAN interface is accessed through the first 10G port which is a shared port for both remote management and standard 10G Ethernet to the host. The HECI (Host Embedded Controller Interface) is used for direct access from the host OS and requires driver support (Linux is currently supported).

The IE resides on the SoC which is on the COM-e module. The Grizzly platform also makes use of a separate management controller on the baseboard. This controller is an ARM-based MCU that communicates with the IE over an IPMB/I²C bus. It maintains several features that are baseboard-specific such as voltage sensors and fan control.

Block Diagram

This block diagram shows the basic configuration of Remote Management on the Grizzly.

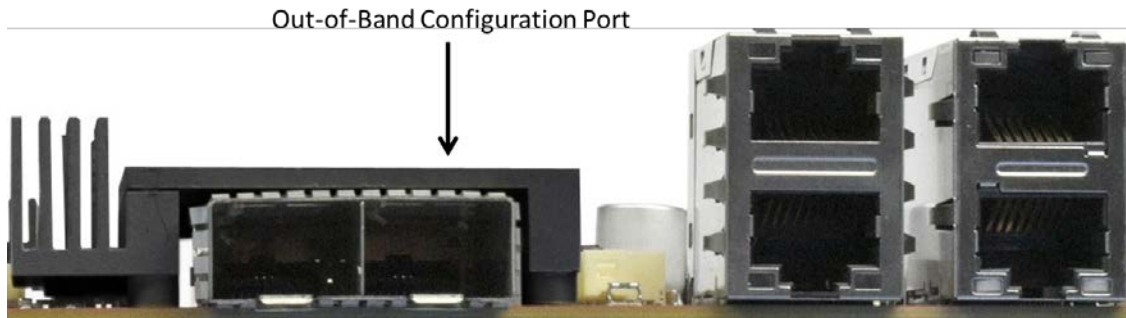
Figure 1. ESU-5070 Remote Management Configuration.



Remote Management Configuration

The 10G SFP shared Management port is the only port capable of out-of-band remote management. The link will always be active and the IE accessible even when the board is in the S5 (shutdown) state.

Figure 2. Out-of-Band Configuration Port



The IE is configured with the default Static IP address 10.99.99.60. Configuration of remote management can be accomplished using BIOS setup or using IPMI commands over the LAN or HECI interfaces.

The current LAN settings for the IPMI interface can be displayed either in BIOS setup or using the “lan print” IPMI command:

```
# ipmitool lan print 1
Set in Progress : Set Complete
Auth Type Support: MD5 PASSWORD
Auth Type Enable : Callback: MD5 PASSWORD
                  : User: MD5 PASSWORD
                  : Operator: MD5 PASSWORD
                  : Admin: MD5 PASSWORD
                  : OEM:
IP Address Source: Static Address
IP Address       : 10.99.99.60
Subnet Mask      : 255.255.255.0
MAC Address      : 00:53:b4:d4:a7:73
SNMP Community String: public
IP Header        : TTL=0x40 Flags=0x40 Precedence=0x00
                  : TOS=0x10
BMC ARP Control  : ARP Responses Enabled, Gratuitous ARP
```

```

Disabled
Gratituous ARP Intrvl: 2.0 seconds
Default Gateway IP: 0.0.0.0
Default Gateway MAC: 00:00:00:00:00:00
Backup Gateway IP: 0.0.0.0
Backup Gateway MAC: 00:00:00:00:00:00
802.1q VLAN ID   : Disabled
802.1q VLAN Priority: 0
RMCP+ Cipher Suites: 1,2,3,6,7,8,11,12,15,16,17
Cipher Suite Priv Max: aaaaaaaaaaXXXX
                        :   X=Cipher Suite Unused
                        :   c=CALLBACK
                        :   u=USER
                        :   o=OPERATOR
                        :   a=ADMIN
                        :   O=OEM
Bad Password Threshold: Not Available

```

Any of the LAN parameters can be modified with the “lan set” command. For example, to change the static IP address, the following would be used:

```
# ipmitool lan set 1 ipaddr <x.x.x.x>
```

Ipmitool

Note: The ipmitool is available at [ipmitool](http://ipmitool.sourceforge.net/).

To access the IPMI functionality on Grizzly, a customized version of the ipmitool utility is used. The standard open-source ipmitool or other IPMI utilities can be used for most functions but to use the HECI interface or to use firmware update commands over IPMI, the customized ipmitool version from VersaLogic needs to be used.

ipmitool can be used for either the HECI interface (locally) or the LAN interface (remotely). Below are some examples of the differences in the options to use based on which interface is being used:

Access the IE using HECI interface locally on host OS:

```
ipmitool mc info
```

Access the IE using LAN interface from remote system connecting to 10G Management Port:

```
ipmitool -I lan -H <hostIP> -U root -P root mc info
```

Accessing the MCU Management Controller

The MCU Management Controller on the baseboard communicates with IE using the IPMB (Intelligent Platform Management Bus) messaging protocol over I²C. To access the MCU using ipmi commands through the IE, messages need to be bridged to the IPMB address of the MCU (0x4E). To accomplish this, the IE supports the SendMessage command which allows for bridging commands to a separate IPMB address. The IE, like any BMC, is always located at IPMB address 0x20.

So when sending an IPMI command to the IE using ipmitool over either the LAN or HECI interface, the address of the remote IPMB needs to be specified to be able to communicate with the MCU. For example:

Access the MCU using HECI interface:

```
ipmitool -t 0x4e mc info
```

Using Serial Over LAN

Serial over LAN enables a remote user to establish a serial console over the LAN interface that will allow access to the BIOS and Host OS on the Grizzly in a similar way to a local serial port using console redirection. To establish an IPMI SOL session to the Grizzly the following is needed:

- A remote system using ipmitool with RMCP+ support
- A LAN connection to the 10G Management interface on Grizzly (IP address)
- The Grizzly BIOS configured so console redirection is enabled on the UART used for SOL
- OS configured to redirect console to the SOL port (in Linux, the SOL device is /dev/ttyS4)

Using ipmitool on the remote session, review the SOL parameters using the ipmitool lanplus interface:

```
ipmitool -I lanplus -H <ipaddr> -U root -P root sol info
```

Establish the SOL session using ipmitool lanplus interface:

```
ipmitool -I lanplus -H <ipaddr> -U root -P root sol  
activate
```

The session will stay open until it loses connection or the following two characters are entered:

```
~.
```

IPMI Commands Supported by IE

The Grizzly remote management is IPMI 2.0 compliant and supports several features including SEL, Sensor events, and Serial over LAN. It does not support a web interface or a virtual KVM. The following IPMI commands are supported and can be sent using `ipmitool` over the HECI or LAN interface. Refer to the IPMI 2.0 Specification for more detail on each command.

Global

- Supported Global IPMI commands:
- Get Device ID
- Cold Reset
- Get Self Test Results
- Set ACPI Power State
- Get ACPI Power State
- Get Device GUID

IPMITOOL EXAMPLES

Get the board's Device ID and other information"

```
ipmitool mc info
```

Chassis

Supported Chassis IPMI commands:

- Get Chassis Capabilities
- Get Chassis Status
- Chassis Control
- Chassis Identify
- Set Chassis Capabilities
- Set Power Restore Policy
- Set System Boot Options
- Get System Boot Options
- Get POH Counter

IPMITOOL EXAMPLES

Get the board's power status:

```
ipmitool chassis power status
```

Power off the board

```
ipmitool chassis power off
```

Power on the board

```
ipmitool chassis power on
```

Watchdog Timer

Supported BMC Watchdog IPMI Commands:

- Reset Watchdog Timer
- Set Watchdog Timer
- Get Watchdog Timer

IPMITOOL EXAMPLES

Get the current watchdog settings:

```
ipmitool mc watchdog get
```

Reset the watchdog to start it counting down:

```
ipmitool mc watchdog reset
```

Shut off the watchdog to stop it from counting down:

```
ipmitool mc watchdog off
```

Device and Messaging

Supported Device and Messaging Commands:

- Set BMC Global Enables
- Get BMC Global Enables
- Clear Message Flags
- Get Message Flags
- Get Message
- Send Message
- Read Event Message Buffer
- Get System GUID
- Get Channel Authentication Capabilities

- Get Session Challenge
- Activate Session
- Set Session Privilege Level
- Close Session
- Get Session Info
- Set Channel Access
- Get Channel Access
- Get Channel Info
- Set User Access
- Get User Access
- Set User Name
- Get User Name
- Set User Password
- Activate Payload
- Deactivate Payload
- Get Payload Activation Status
- Get Payload Instance Info
- Set User Payload Access
- Get User Payload Access
- Get Channel Payload Support
- Get Channel Payload Version
- Master Write-Read

Event

Supported Event IPMI Commands:

- Get Event Receiver
- Platform Event (a.k.a. “Event Message”)

IPMITOOL EXAMPLES

Send an example platform event message:

```
ipmitool event 1
```

PEF and Alerting

Supported PEF and Alerting Commands:

- Get PEF Capabilities
- Arm PEF Postpone Timer
- Set PEF Configuration Parameters
- Get PEF Configuration Parameters
- Set Last Processed Event ID
- Get Last Processed Event ID
- Alert Immediate
- PET Acknowledge

FRU

The MCU and IE FRU Data contains useful information on a board that includes serial numbers, product codes, and builds dates. This can be useful for inventory management.

MCU FRU DATA

The MCU contains FRU data that is read only.

Supported FRU IPMI commands:

- Get FRU Inventory Area Info
- Read FRU Data

IPMITOOL EXAMPLES

Print FRU data contained in the MCU

```
ipmitool -t 0x4e fru print
FRU Device Description : Builtin FRU Device (ID 0)
Board Mfg Date        : Mon Jan 13 00:00:00 2020
Board Mfg              : VersaLogic
Board Product         : Grizzly-Carrier
Board Part Number     : A-BBe-70
Product Manufacturer   : VersaLogic
Product Name          : Grizzly
Product Part Number    : EPU-5070
Product Version        : v1.0
Product Asset Tag      : Generic
```

IE FRU DATA

New FRU data can be generated and loaded onto the IE with custom data if desired.

Supported FRU IPMI commands:

- Get FRU Inventory Area Info
- Read FRU Data
- Write FRU Data

IPMITOOL EXAMPLES

Print FRU data contained in the IE:

```
ipmitool fru print
```

FRU Device Description : Builtin FRU Device (ID 0)

```
Board Mfg Date       : Mon Jul 10 16:30:00 2017
Board Mfg            : VersaLogic
Board Product        : COMe-50
Board Serial         : 922223
Board Part Number    : A-COMe-50EC Rev 1.00
Product Manufacturer : VersaLogic
Product Name         : Grizzly
Product Part Number  : VL-ESU-5070ECP-00
```

SEL

The IE implements a standard System Event Log (SEL). The SEL stores timestamped events in the system such as boot events and sensor threshold assertions.

Supported SEL IPMI commands:

- Get SEL Info
- Get SEL Allocation Info
- Reserve SEL
- Get SEL Entry
- Add SEL Entry
- Partial Add SEL Entry
- Clear SEL
- Get SEL Time

IPMITOOL EXAMPLES

List all SEL entries:

```
ipmitool sel list
```

Get SEL Info:

```
ipmitool sel info
```

Clear SEL:

```
ipmitool sel clear
```

Sensor

Supported Sensor IPMI commands:

- Set Sensor Hysteresis
- Get Sensor Hysteresis
- Set Sensor Threshold
- Get Sensor Threshold
- Set Sensor Event Enable
- Get Sensor Event Enable
- Re-arm Sensor Events
- Get Sensor Event Status
- Get Sensor Reading

IPMITOOL EXAMPLES

List all sensor values and thresholds:

```
ipmitool sensor
```

SDR

Supported SDR IPMI commands:

- Get SDR Repository Info
- Get SDR Repository Allocation Info
- Reserve SDR Repository
- Get SDR
- Add SDR
- Partial Add SDR
- Delete SDR

- Clear SDR Repository
- Get SDR Repository Time
- Run Initialization Agent

IPMITOOL EXAMPLES

Get SDR repository information:

```
ipmitool sdr info
```

LAN

Supported LAN IPMI commands:

- Set LAN Configuration Parameters
- Get LAN Configuration Parameters

IPMITOOL EXAMPLES

Get LAN parameters for LAN channel 1 (the IE remote management channel):

```
ipmitool lan print 1
```

Serial

Supported Serial IPMI commands:

- Set Serial/Modem Configuration
- Get Serial/Modem Configuration
- Set SOL Configuration Parameters
- Get SOL Configuration Parameters

IPMI Commands Supported by MCU

IPMI commands bridged to the MCU on the Grizzly baseboard using IPMB address 0x4e are limited mainly to Sensor, FRU, and SDR commands.

Global

Supported Global IPMI commands:

- Get Device ID
- Cold Reset

Event

Supported Event IPMI Commands:

- Set Event Receiver
- Get Event Receiver

FRU

Supported FRU IPMI commands:

- Get FRU Inventory Area Info
- Read FRU Data
- Write FRU Data

Sensor

Supported Sensor IPMI commands:

- Get Sensor Hysteresis
- Get Sensor Threshold
- Get Sensor Reading

SDR

Supported SDR IPMI commands:

- Get SDR Repository Info
- Reserve SDR Repository
- Get SDR

Sensor Type Details

Detailed information regarding sensors.

IE Sensor

Sensor ID : CPU0 DIMM0 Temp (0x11)

Entity ID : 8.1

Sensor Type (Threshold): Temperature

Lower Non-Recoverable: -40.000

Lower Critical : -35.000

Lower Non-Critical: -30.000

Upper Non-Critical: 85.000

Upper Critical : 90.000

Upper Non-Recoverable: 95.000

Positive Hysteresis: 2.000

Negative Hysteresis: 2.000

Assertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Deassertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Sensor ID : CPU0 DIMM1 Temp (0x12)

Entity ID : 8.1

Sensor Type (Threshold): Temperature

Lower Non-Recoverable: -40.000

Lower Critical : -35.000

Lower Non-Critical: -30.000

Upper Non-Critical: 85.000

Upper Critical : 90.000

Upper Non-Recoverable: 95.000

Positive Hysteresis: 2.000

Negative Hysteresis: 2.000

Assertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Deassertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Sensor ID : CPU0 DIMM2 Temp (0x13)

Entity ID : 8.1

Sensor Type (Threshold): Temperature

Lower Non-Recoverable: -40.000

Lower Critical : -35.000

Lower Non-Critical: -30.000

Upper Non-Critical: 85.000

Upper Critical : 90.000

Upper Non-Recoverable: 95.000

Positive Hysteresis: 2.000

Negative Hysteresis: 2.000

Assertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Deassertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Sensor ID : CPU0 DIMM3 Temp (0x14)

Entity ID : 8.1

Sensor Type (Threshold): Temperature

Lower Non-Recoverable: -40.000

Lower Critical : -35.000

Lower Non-Critical: -30.000

Upper Non-Critical: 85.000

Upper Critical : 90.000

Upper Non-Recoverable: 95.000

Positive Hysteresis: 2.000

Negative Hysteresis: 2.000

Assertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Deassertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Sensor ID : CPU0 Temp (0x10)

Entity ID : 3.1

Sensor Type (Threshold): Temperature

Lower Non-Recoverable: -40.000

Lower Critical : -35.000

Lower Non-Critical: -30.000

Upper Non-Critical: 73.000

Upper Critical : 78.000

Upper Non-Recoverable: 83.000

Positive Hysteresis: 2.000

Negative Hysteresis: 2.000

Assertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Deassertions Enabled: Inc- Inc+ lcr- lcr+ lnr- lnr+ unc- unc+ ucr- ucr+ unr- unr+

Sensor ID : IPMI Watchdog (0x3)

Entity ID : 7.1

Sensor Type (Discrete): Watchdog2

Sensor ID : System Event Log (0x7)

Entity ID : 7.1

Sensor Type (Discrete): Event Logging Disabled

MCU Sensor (-t 0x4e)

Sensor ID : V_POWER_IN (0x1)

Entity ID : 193.111

Sensor Type (Threshold): Voltage

Lower Non-Recoverable: 10.880

Lower Critical : 11.136

Lower Non-Critical: 11.520

Upper Non-Critical: 30.080

Upper Critical : 30.336

Upper Non-Recoverable: 30.720

Positive Hysteresis: 0.256

Negative Hysteresis: 0.256

Sensor ID : V12 (0x2)
Entity ID : 193.111
Sensor Type (Threshold): Voltage
Lower Non-Recoverable: 10.880
Lower Critical : 11.200
Lower Non-Critical: 11.520
Upper Non-Critical: 12.480
Upper Critical : 12.800
Upper Non-Recoverable: 13.120
Positive Hysteresis: 0.128
Negative Hysteresis: 0.128

Sensor ID : V5_SBY (0x3)
Entity ID : 193.111
Sensor Type (Threshold): Voltage
Lower Non-Recoverable: 3.840
Lower Critical : 3.968
Lower Non-Critical: 4.096
Upper Non-Critical: 5.504
Upper Critical : 5.632
Upper Non-Recoverable: 5.760
Positive Hysteresis: 0.128
Negative Hysteresis: 0.128

Sensor ID : V12_S0 (0x4)
Entity ID : 193.111
Sensor Type (Threshold): Voltage
Lower Non-Recoverable: 10.880
Lower Critical : 11.200
Lower Non-Critical: 11.520
Upper Non-Critical: 12.480
Upper Critical : 12.800
Upper Non-Recoverable: 13.120

Positive Hysteresis: 0.128
Negative Hysteresis: 0.128

Sensor ID : V3P3 (0x5)
Entity ID : 193.111
Sensor Type (Threshold): Voltage
Lower Non-Recoverable: 2.688
Lower Critical : 2.880
Lower Non-Critical: 3.072
Upper Non-Critical: 3.520
Upper Critical : 3.776
Upper Non-Recoverable: 3.968
Positive Hysteresis: 0.128
Negative Hysteresis: 0.128

Sensor ID : V1P5 (0x6)
Entity ID : 193.111
Sensor Type (Threshold): Voltage
Lower Non-Recoverable: 1.280
Lower Critical : 1.344
Lower Non-Critical: 1.408
Upper Non-Critical: 1.792
Upper Critical : 1.856
Upper Non-Recoverable: 1.920
Positive Hysteresis: 0.128
Negative Hysteresis: 0.128

Sensor ID : V0P9 (0x7)
Entity ID : 193.111
Sensor Type (Threshold): Voltage
Lower Non-Recoverable: 0.640
Lower Critical : 0.704
Lower Non-Critical: 0.768
Upper Non-Critical: 1.088

Upper Critical : 1.152
Upper Non-Recoverable: 1.216
Positive Hysteresis: 0.128
Negative Hysteresis: 0.128

Sensor ID : V1P8 (0x8)
Entity ID : 193.111
Sensor Type (Threshold): Voltage
Lower Non-Recoverable: 1.408
Lower Critical : 1.472
Lower Non-Critical: 1.536
Upper Non-Critical: 2.048
Upper Critical : 2.176
Upper Non-Recoverable: 2.304
Positive Hysteresis: 0.128
Negative Hysteresis: 0.128

Sensor ID : BaseBoard Temp (0x9)
Entity ID : 193.111
Sensor Type (Threshold): Temperature
Lower Non-Recoverable: -45.000
Lower Critical : -40.000
Lower Non-Critical: -35.000
Upper Non-Critical: 80.000
Upper Critical : 85.000
Upper Non-Recoverable: 90.000
Positive Hysteresis: 2.000
Negative Hysteresis: 2.000

Sensor ID : COM Board Temp (0xa)
Entity ID : 193.111
Sensor Type (Threshold): Temperature
Lower Non-Recoverable: -45.000
Lower Critical : -40.000

Sensor Type Details

Lower Non-Critical: -35.000
Upper Non-Critical: 80.000
Upper Critical : 85.000
Upper Non-Recoverable: 90.000
Positive Hysteresis: 2.000
Negative Hysteresis: 2.000

Sensor ID : Fan Tach (0xb)
Entity ID : 29.111
Sensor Type (Threshold): Fan
Lower Non-Recoverable: 0.000
Lower Critical: 320.000
Lower Non-Critical: 640.000
Upper Non-Critical: 4800.000
Upper Critical : 5440.000
Upper Non-Recoverable: 6080.000
Positive Hysteresis: 128.000
Negative Hysteresis: 128.000